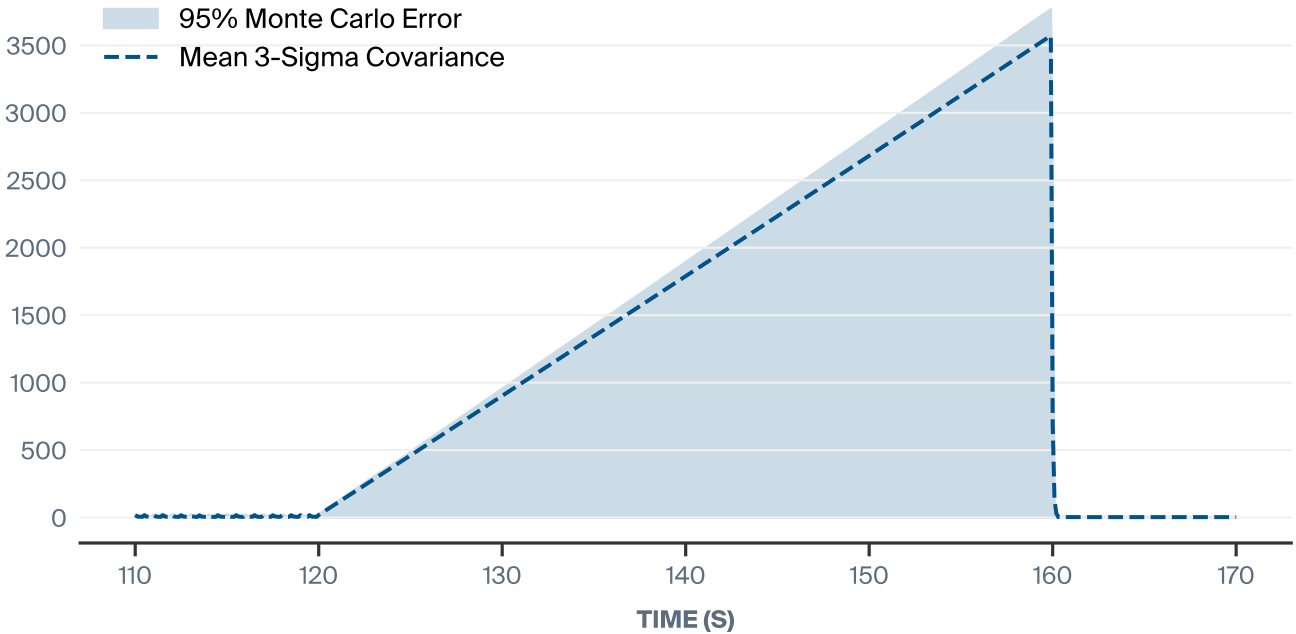BLACKOUT POSITION ERROR ENVELOPE